Advanced Educational Activities Pvt. Ltd.

(Key and Solutions for AIMCAT2412)

Key

SECTION – I

1. D	7. C	13. C	19. D
2. B	8. D	14. B	20. 3412
3. D	9. D	15. A	21. B
4. D	10. D	16. D	22. C
5. B	11. C	17. 2413	23. A
6. B	12. A	18. 1423	24. A

SECTION – II

1. 783649512	6. 6	11. A	16. A
2. 287649315	7. D	12. A	17. 10
3. B	8. B	13. B	18. B
4. 64951382	9. D	14. A	19. 250
5. D	10. A	15. C	20. C

SECTION – III

1. C	7. 50	13. C	19. C
2. A	8. A	14. B	20. 0
3. D	9. 18	15. A	21. 26
4. 0.04	10. C	16. 512	22. D
5. A	11. B	17. D	
6. B	12. –3	18. 120	

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 519

1. Consider these statements: *'In those cases, **it would be ideal to not have any recombination but to have the things that go well together stuck together for good**. To understand why that might be advantageous, think about doing laundry.'* So, the author uses the laundry example to explain why it would be advantageous/ideal to not have recombination.
Option A: The example is to show how recombination is not always useful, and the reverse might be useful i.e. avoiding recombination. The example is not to show that recombination is 'disastrous', which is an exaggeration. Option A is not the answer.
Option B: Recombination is counterproductive in some cases. This is understood from the example, yes. But the example doesn't talk about heterogeneity. The example is about how it may be ideal that similar genes that go well together are stuck together. The discussion doesn't bring in the topic of heterogeneity or similarity of genes which is not relevant here. Option B is not the answer.
Option C: The focus of the example is not on keeping specific combinations of genes apart. It is on keeping some combinations together and avoiding mixing up of genes. It is about treating a bundle of genes as one unit because of the benefits they provide. Option C is not the answer.
Option D: This is the point of the example, that a bundle of genes that do well together shouldn't be reshuffled and kept together. That continues giving us the benefits of that group as a whole rather than creating new groups. Option D is the answer. Choice (D)

2. Option A: Supergenes arise from mutations in genes locked together by inversion. The last para raises the question of how the inversion arises. From an inverted fragment of DNA, mutations that make it beneficial will result in a supergene.

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However, the passage does not mention whether the formation of a supergene is accidental or if it is in response to an environmental trigger (or any other way). Hence, we cannot conclude from the passage that a supergene is formed purely accidentally. Option A is not the answer.
Option B: This can be understood from '***Any beneficial mutations arising within the inversion could then promote its spread as a supergene.***' A bundle of genes with a mutation that offers evolutionary advantages is propagated in the form of a supergene. Option B is the answer.
Option C: If the inversion doesn't offer any evolutionary benefits, then it won't be propagated, nor can it be referred to as a supergene, in the first place. So, the idea that the 'supergene is stifled' is incorrect. Option C is not the answer.
Option D: A supergene is not necessarily just a big chunk of gene, locked in an inversion. If a bundle offers evolutionary advantages, then it will be propagated as a supergene. It is not true that a 'supergene is propagated whenever it contains a chunk of genes locked together'. Hence, Option D is not the answer. Choice (B)

3. Consider the sentences: '***That fact may help to explain why complex vital traits aren't routinely secured as supergenes: Ordinary selection pressures are often sufficient to preserve the traits.***'
Option A: A suite of genes being well-adapted to its surroundings doesn't necessarily prevent its inversion for sure. This can be understood from '***If a suite of genes is already well adapted to its surroundings, locking it into an inversion will not suddenly allow it to take off as a supergene.***' So, such an inversion won't be propagated, but it doesn't mean the inversion cannot happen at all. Option A is not the answer.
Option B: Selection pressures are sufficient to preserve complex vital traits because those complex traits have evolutionary benefits. However, it is supergenes where the traits are preserved by the prevention of recombination, not in the normal suite of genes preserved because of selection pressures. Hence, Option B is not the answer.
Option C: Ordinary selection has been mentioned as a factor the protects a well-adapted suite of genes even when not locked in an inversion. This option contradicts that information. Hence, Option C is not the answer.
Option D: The author says that complex vital traits aren't secured as supergenes because the genes that are well-adapted to the surroundings are already preserved to make sure they are passed on without tampering. So, an inversion is not the only way to secure/protect them. They are propagated as it is because of the evolutionary benefits the bundle of genes provides. Option D is the answer. Choice (D)

4. Option A: Inversions are not propagated when they don't provide an evolutionary benefit, according to the author. Similarly, we cannot say that inversions happen only when there is an evolutionary benefit as understood from "The question of whether an adaptation precedes an inversion or vice versa, Lotterhos realized, might never be answerable" This option can be inferred. Option A is not the answer.
Option B: This can be understood from the passage, from the following sentence: '***Any beneficial mutations arising within the inversion could then promote its spread as a supergene.***' When an inversion has a mutation that is beneficial, it may take off as a supergene. Option B is not the answer.
Option C: Consider this: 'A larger inverted fragment of DNA was more likely to capture multiple genes and lock them together as a single entity. Any beneficial mutations arising within the inversion could then promote its spread as a supergene.' It implies that larger the inverted fragment, higher the chance of a beneficial mutation. Also, if there is a mutation, the locked inversion may take off as a supergene. Option C is not the answer.
Option D: This cannot be inferred from the passage. Not all inversions produce beneficial mutations, and hence not all inversions necessarily give us supergenes. Option D is the answer. Choice (D)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 538

5. Option A: The passage doesn't really talk about individual novels or their quality. In fact, there is hardly any discussion about individual titles or what they entail. It is about the overall state of literature, and novels. Hence, Option A is not the answer.
Option B: This is apt, since the author mentions how literature's epilogue is being written, that we are only trying to conjure a ghost, that what we have now in the form of literature or as a replacement is kitsch. Hence, Option B is the answer.
Option C: The chief argument of the passage is not about the popularity of literature but the overall quality of literature and whether we even have something that is the equivalent of literature. Hence, Option C is not the answer.
Option D: While the author subtly touches upon the quality of modern-day writers, it doesn't take up too much space. The passage overall doesn't extensively delve upon whether or not modern-day writers are putting in any effort. Hence, Option D is not the answer. Choice (B)

6. Consider the sentence: '***we shall go on for quite a long time talking of books and writing books, pretending all the while not to notice that the church is empty and the parishioners have gone elsewhere to attend other gods.***'

Option A: While the church and parishioners are metaphors for novels and novel-readers/purveyors of art in general, the expression overall is not a metaphor for the decline in ***the quality of the novel***. It is an analogy/parallel example to demonstrate how the ***popularity of the novel***, or rather the following has subsided. Option A is not the answer.
Option B: This is apt since the expression provides a parallel case to illustrate what is the current state of readership of the novel. There is no one in the church and the people have left for elsewhere: this expresses the author's opinion about the present/contemporary readership of the novel. The novel has lost ground to other forms of art. Hence, Option B is the answer.
Option C: While the expression tried to illustrate that literature has lost its following, it is not an example, per se. It is a parallel situation to dramatize what is happening with respect to the novel. Hence, Option C is not the answer.
Option D: The analogy doesn't really point to other artforms. It doesn't 'compare' popularities. All it says is that followers of the novel have gone elsewhere. Option D is not the answer. Choice (B)

7. The two things can be understood from here: Since then, two things have happened. ***The novel*** – which was meant to fuse poetry and philosophy, to subsume all other genres and even the entire universe – ***has been reduced to "literary fiction"***: a genre that approaches writing as if the 20th century had never happened. At the same time, ***the digital age has taken information overload to a whole new level***.
Option A: The novel has not subsumed all other genres. It was expected to, but it went the opposite way, according to the sentences above. Option A is not the answer.
Option B: While it is partially true that the novel is not complex, or rather sophisticated enough now to subsume poetry and philosophy the way it was expected to, it is not "despite" the overload of information. These are two independent developments: the novel going down in standards, and the advent of digital age bringing an overload of information. Option B is not the answer.
Option C: This is apt as it represents the correlation of two things that have taken place since the 90s. Firstly, the novel didn't evolve, but was rather "reduced to" just literary fiction. Secondly, the digital age has brought about an overload of information. Hence, Option C is the answer
Option D: While this explains the author's purpose behind using the term "reduced to literary fiction", this doesn't talk about the advent of digital age and its consequences. Option D is not the answer. Choice (C)

8. We are looking for an expression that doesn't directly equate literature to a poorer/worse form of itself, as understood from the passage.
Option A: This was used here: 'Literature only survives as literary-fiction kitsch: a "parody of past forms"; a "pantomime of itself".' Literature has been called a parody of past forms to show how it doesn't carry the weight, seriousness or quality that it probably used to. Hence, Option A is not the answer.
Option B: The novel has been called both 'a parody of past forms' (a poor remaking of the original not intended to be taken seriously) and 'a pantomime of itself' (a joke on itself of sorts) to represent the lowering of the quality. Option B is not the answer.
Option C: From '***We are no longer writing literature's conclusion but its "epilogue": ours is a "literature which comes after literature"***', we can understand that the standard of literature in the current times is not even its conclusion, rather something that comes at the end of a work of art. So, what we are writing now, the literature after literature, is just a shadow/ghost of what it used to be. Hence, Option C is not the answer.
Option D: Consider this: '***In "The Literature of Exhaustion", Barth had envisaged how the "felt ultimacies of our time" could become the material of future works.***' The expression 'felt ultimacies' does not represent the debilitated state of literature. It is what Barth feels would be the inspiration for future works. It can be inferred to be something less desirable, but it doesn't in any way indicate a debilitated state when compared to other commentaries. That's why the next sentence reads 'Iyer cranks this up a notch.' Option D is the answer. Choice (D)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 507

9. There have been three different periods of focus areas: pre-1960s, 1960s, and 1970s.
Option A: This was the focus area during the pre-1960s as understood from '***Pre-1960s a great deal of environmental concern was focused upon small-scale localized problems.***' Small-scale localised problems indicates issues that are restricted to one area without impacting others and, hence, not global. Option A is not the answer.
Option B: Consider this: ***In the 1960s concern grew about the interrelation between human actions and more complex environmental implications.*** So, this was the period during which the focus was on how human actions were related to environmental problems, and as an extension, how human actions affected the environment. Option B is not the answer.
Option C: This can be understood from: ***By the early 1970s there was recognition not just of the interconnectedness of environmental issues, and of the problems of cumulative damage, but also that these were global problems that needed internationally***

coordinated responses. So, by 1970s, the focus area was to coordinate international responses to global environmental problems which were all understood to be interconnected. Option C is not the answer.
Option D: The author clearly argues that we cannot really establish a causal relationship between environmental problems and overpopulation. So, it's clearly not a focus area. The author discusses the neo-Malthusian theory to highlight that growing concerns about overpopulation affecting the environment are misplaced. Option D is the answer. Choice (D)

10. Option A: Neo-Malthusian explanations according to the author are oversimplified and misguided. The explanations don't 'oversimplify' the evidence. This is a misrepresentation of the given ideas. Option A is not the answer.
Option B: It is true that population growth and land degradation are linked but the relationship between them is too complex for one rule to be established. In fact, the author says that we cannot attribute land degradation entirely to just one parameter like population growth. So, while we can't say that there is a fixed causal relationship between land degradation and population size, we can't say that there is no causal relationship either. Hence, Option B is not the answer.
Option C: Linking land degradation "only" to population size ***masks the complexity between population dynamics and labour-land relationships.*** The author doesn't say that the complexity of population dynamics is the reason why land degradation cannot be linked to a 'single parameter like' population size. Option C is not the answer.
Option D: That studies have not been able to show that there is a clear causal relationship between land degradation and population size can be understood from: '***The misguided simplicity of*** neo-Malthusian explanations of environmental problems ***is shown in studies that have tried to*** explain land degradation in terms of population dynamics.' The evidence is contradictory for such relationships. So, we can understand that no studies have managed to prove this. Option D is the answer. Choice (D)

11. Option A: Systemic changes do not necessarily combine to give us cumulative changes. If that were the case, it wouldn't be a duality anymore. Systemic change and cumulative change are two different types of change analysed by environmental scientists. Option A is not the answer.
Option B: Even cumulative change takes place at a global scale as understood from '***Cumulative change, on the other hand, occurs when many discrete events become significant because their distribution is global.***' Option B is not the answer.
Option C: The author says that '***systemic change occurs when there is a direct impact on a physically interconnected, global system***' and '***Cumulative change, on the other hand, occurs when many discrete events become significant because their distribution is global.***' Hence, Option C is the answer.
Option D: The author doesn't say that cumulative change happens when the events are discrete. Cumulative change occurs when discrete events become significant because of ***their global distribution***. Similarly, systemic change doesn't happen when systems are interconnected. It occurs when one interconnected global system is impacted. Hence, Option D is not the answer. Choice (C)

12. Consider the sentences: 'Cumulative change ... occurs when many discrete events become significant because their distribution is global or because, added together, ***their impact is felt across a large proportion of the globe***. For example, across the Americas, Africa and Asia-Pacific, ***the relatively small amounts of tropical forest lost in each area*** being converted from forest to farmland ... ***add up to a global-scale problem*** affecting humid tropical forests...'
Option A: The author talks about the small amounts of tropical forest lost in each area to show how those small amounts can add up to a global scale problem. This is a good example of cumulative change. Option A is the answer.
Option B: The example of small amounts of tropical forest was not to establish the importance of these forests. That is already known. The example was to show the impact of small events when combined together. Option B is not the answer.
Option C: While this is true, it doesn't come from the example of loss of small amounts of tropical forest. The 'loss' in particular is directed towards explaining the global impact of these small events (event here is the loss of small areas of forest). Option C is not the answer.
Option D: This is an advice the author provides to those working on cumulative changes for effectiveness. However, it doesn't explain why the small amounts of tropical forests being lost was mentioned or what that example establishes. Hence, Option D is not the answer. Choice (A)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 530

13. Option A: Consider this: '***Just as research has shown that feedback from caregivers shapes infant productions of sounds, feedback from the discriminator network shapes the sound productions of the generator network.***' So, the sentence says the two processes are similar, in the sense that the feedback moulds the sounds produced. However, from this we cannot infer whether the feedback mechanism of discriminator is like that of the caregiver or is similar to a

certain extent. The mechanisms could be entirely different. Option A is not the answer.
Option B: The feedback is provided by the discriminator to the generator and not the other way around, across several iterations. Hence, Option B is not the answer.
Option C: Consider this: '***The researchers then recorded the processing activity of the neural network, focusing on the artificial neurons in the layer of the network where sounds are first analyzed (to mirror the brainstem readings). It was these signals that closely matched the human brain waves.***' Such a comparison with human brain waves is only possible if neural network processing activity can be compared. Hence, Option C is the answer.
Option D: The passage only discusses comparisons between the sound produced by the generator and the training sample. We don't know if networks can be trained to differentiate between two languages, leave alone identify the structural/syntactical differences between two languages. It may be possible, but we cannot infer it from the passage. Option D is not the answer. Choice (C)

14. Consider this: '***In this study, the discriminator was initially trained on a collection of either English or Spanish sounds. Then the generator had to find a way of producing them. As a result of this training, the discriminator also got better at distinguishing between real and generated ones.***'
Option A: The passage argues that the processing of sounds by artificial neural networks and brains can be similar. The passage doesn't delve into the differences in processing of artificial and natural sounds. Also, the given statement only discusses that the discriminator can differentiate between real and artificial sounds. How it does that cannot be concluded. Hence, Option A cannot be concluded from the statement.
Option B: If the given statement is true, that the discriminator got better at distinguishing between real and generated sounds, it is also implied that it is possible to accurately differentiate between artificial and natural sounds. Hence, Option B is the answer.
Option C: The way these sounds are processed is similar in case of artificial neural networks and brains. However, neither from the statement nor from the passage can we understand that the processing is different in case of natural and artificial sounds. Hence, Option C is not the answer.
Option D: We cannot understand from the given statement whether artificial neural networks change their processing based on the sound they have to process. Hence, Option D is not the answer. Choice (B)

15. Option A: Consider the statements: '***The brain waves showed that the English- and Spanish-speaking participants heard the "bah" sound differently... The brainstem of English speakers responds to the "bah" sound slightly earlier than the brainstem of Spanish speakers, and the GAN trained in English responded to that same sound slightly earlier than the Spanish-trained model.***' The parallel mentioned here is that both English speakers and GAN trained in English respond to the same sound 'bah' slightly earlier than Spanish speakers and Spanish-trained GAN. Hence, Option A is the answer.
Option B: While the GAN models differ in their response time the same way as participants, we don't know whether the delay is the same amount or any different. Hence, Option B is not the answer.
Option C: The models don't differ in their responses. They differ in their response times, which is the parallel the author is drawing between GAN and the human brains. Option C is not the answer.
Option D: There is a difference in response times of English and Spanish-trained GAN (just as there is in case of English and Spanish speakers). Whether GANs take the same or more amounts of time as the English speakers hasn't been discussed. Hence, Option D is not the answer. Choice (A)

16. Because artificial neural networks process sounds the same way as brains do, the author believes that artificial neural networks can be trained, which means brains don't have an inherent hardware or inbuilt mechanism. Children learn to do it the way artificial neural networks are trained.
Option A: If true, this option doesn't give us any further insight into whether brains have in-built capabilities to process languages at the time of human birth. Option A is not the answer.
Option B: Even if true, this option doesn't throw light on the original capabilities of the brain and whether it comes equipped with hardware or software for processing languages. Option B is not the answer.
Option C: Even if true, it doesn't tell us whether brains come already with hardware or software. In other words, this statement can be true irrespective of whether brains come with hardware or software already built in to process languages. Option C is not the answer.
Option D: If true, this strengthens the argument that brains don't come already equipped with hardware or software to process languages. They must be trained during infanthood, which is why their processing can be mimicked by artificial neural networks which can be trained in a similar way. Option D is the answer. Choice (D)

Solutions for questions 17 and 18:

17. Sentence 1 is grammatically dependent on another sentence that provides context for 'this'. Plus, this sentence has a connector 'also' which means something else has been discussed prior to this.

Sentence 2 is grammatically and contextually independent. It is also quite vague, making it a good candidate for being the first sentence of the paragraph.
Sentence 3 refers to 'this ethereal essence' and explains how the two branches of chemistry—organic and inorganic—came about.
Sentence 4 is grammatically independent, but introduces the concept of scientists finding things that didn't exist.
3 has to follow 1 because 'this ethereal essence' refers to the 'mysterious elan vital'. Similarly, both 2 and 4 have to come prior to the 13 block, because they are still talking about 18th century scientists trying to discover things or imagine things that don't exist. Between 2 and 4, 2 has to come first because 2 doesn't have a starting point and talks about chemistry's journey across various centuries of various scientists. 4 offers a fixed timeline. So, the answer is 2413. Ans: (2413)

18. All four sentences are technically independent sentences here. So, we have to look at critical reasoning in order to be able to identify the flow of the para.
Sentence 1 says that the causes of war and peace run deep into questions like 'why armies form'. So, this sentence establishes context for any sentence that talks about why armies form. It also starts with 'to begin with', which makes it the start of the para.
Sentence 2 explains that wars of a continental scale don't happen by instinct and that there is a reason for it.
Sentence 3 explains what lies at the core of wars. So, 3 explains the reason that 2 initiates. The reason that 2 alludes to, is explained by 3. In other words, sentence 2 tells us there is a bigger reason than just instinct for wars. That reason, according to 3 is our capacity to think abstractly and to conceive and to create.
Sentence 4 talks about how only humans indulge in war.
We can understand that both 1 and 4 talk about how there is a deep reason behind why humans indulge in wars. 2 and 3 elaborate on those reasons. 3 follows 2 because it offers the most specifics and connects with 2 with 'in fact'. Between 1 and 4, 1 comes first because 1 introduces the scale of not just war but also peace (none of the other 3 sentences touch upon peace) with the expression 'to begin with'. So, the answer is 1423. Ans: (1423)

Solution for question 19:

19. Option A: The para doesn't talk about manufacturing's "dependency" on technological breakthroughs. It talks about the intimate link between manufacturing and technology, which is why breakthroughs happen more often in this sector. That is what gives manufacturing a higher productivity performance. Option A is not the answer.
Option B: The option mentions that manufacturing produces more technological breakthroughs than any other sector. The para does not make this comparison and only credits manufacturing for all technological breakthroughs ('the other sectors are only capable of passively dealing with technology'). Option B is not the answer.
Option C: While the idea is partially true, this option doesn't mention productivity at all, which is a significant idea in the para. The para tries to connect productivity with technological breakthroughs, both of which are in greater proportions in manufacturing than in other sectors. Option C is not the answer.
Option D: This mentions all the ideas accurately. Manufacturing shows higher productivity performance than other sectors because there is a close relationship between this sector and technology. Option D is the answer. Choice (D)

Solution for question 20:

20. Sentence 1 has a negative connector 'however', explaining how new situations require new responses.
Sentence 2 is an independent sentence about the relation between social evolution, custom and origination.
Sentence 3 talks about how social rather than biological evolution has been witnessed in humans during recorded time.
Sentence 4 talks about what custom and tradition are at a species level and at an individual level.
We start with pairing 4 and 1. 4 talks about how customs and traditions are adjustments to 'frequently repeated' situations. Sentence 1 talks about 'new situations' arise requiring new responses. These two sentences are connected using 'however.' 2 summarises both 4 and 1 by explaining how, social evolution is a combination of the old (custom) and the new (origination). Sentence 3 precedes the 41 block, because it starts with the idea of evolution and narrows it down to how social evolution is more prevalent, which is further explained in 4, 1 and 2. Hence, the order is 3412. Ans: (3412)

Solutions for questions 21 and 22:

21. Option A: The strife between the genders and the strife between the classes are two different ones. The option seems to suggest that there is a strife between the genders and the classes, which is not true. Hence, Option A is not the answer.
Option B: This represents the main idea of the paragraph, that there should be tension between new ideas and the old, that there should be resistance to innovation, and that tension (like the tension between genders or classes, provided as other examples of tension) helps push the whole entity (obviously means society, since we are talking about conservative and radical, genders and classes) forward. Hence, Option B is the answer.

Option C: The passage mentions that innovation before being allowed to enter human race must face objection and opposition. However, the primary idea of the passage is that such a process will help in creating a strong society and will stimulate development. The option doesn't capture this main point. Hence, Option C is not the answer.
Option D: This para misrepresents the idea of innovation. Sometimes, innovation that doesn't pass the resistance will fail. This option seems to suggest that ideas that do not pass the resistance don't actually turn to/bring innovation. Hence, Option D is not the answer. Choice (B)

22. Option A: The author doesn't say that morality is the reason that drives foreign policy. It doesn't say that leaders choose morality over reason. The para says they use morality as an excuse because realism is hard to sell to people. Hence, Option A is not the answer.
Option B: While this option is partly true, it doesn't convey the full idea of the para, specifically the part that talks about how realism is a hard sell, and how it is not openly spoken about by leaders as a reason to pursue power. Option B is not the answer.
Option C: This option explains all the ideas in the para: realism (pure selfishness) is not popular, moral reasons are popular, and, hence, leaders sell war as a moral crusade or ideological contest rather than using realism. Option C is the answer.
Option D: The option seems to suggest what leaders should do. The para doesn't say it is ill-advisable for leaders to admit realism. The para says realism not being popular is why leaders use alternative justifications for war. Hence, Option D is not the answer. Choice (C)

Solutions for questions 23 and 24:

23. The given statement comes in the first blank, followed by elaboration of what that 'gap' is that has been referred to in the first sentence of the para (after blank 1). Option 1 is the answer. It cannot be blank 2 because on either side of the blank, there is a discussion about ethnography that is connected with the word 'then' (after blank 2). It cannot be blank 3 because in the sentence after blank 3 there is 'these gaps' referring to the sentence before blank 3, which means those two statements are connected. It cannot be blank 4 because it doesn't make sense to define what the gap is at the end of the paragraph after gaps have already been discussed. Choice (A)

24. The sentence after blank 2 explains why the defence mechanism of pangolins is what makes them susceptible to capture by traffickers. So, this sentence refers to the first sentence in the given para. So, the bold statement cannot be placed in blank 2. Similarly, the subsequent sentences are all about how pangolins are captured and how poachers harm these harmless creatures. The first part of the bold statement starts with pangolins and their prey, which precedes the idea of the rest of the para which is about how pangolins themselves are hunted. That discussion only makes sense after introducing how pangolins are harmless to any creature. Hence, the bold statement can be best positioned in Blank 1. Option 1 is the answer. Choice (A)

Difficulty level wise summary - Section I

Level of Difficulty	Questions
Very Easy	–
Easy	12
Medium	1, 3, 7, 9, 11, 13, 14, 15, 16, 17, 19, 24
Difficult	2, 4, 5, 6, 8, 10, 21, 22, 23
Very Difficult	18, 20

SECTION – II

Solutions for questions 1 to 5:

To form the largest number, we should try to use all the digits in the grid. It is not possible to use all the digits in the grid if we start at 9. This can be visualized if we shade the grid like a chessboard, as shown the figure.

2	8	7
1	3	6
5	9	4

Irrespective of the place that we start from, we have to alternate between grey and white squares. If we have to form a nine-digit number and if we start with a grey square, we have to end with a grey square (since the total number of digits is odd, the number will be of the form GWGWGWGWG, where G represents a number in a grey square and W represents a number in white square). In that case we will require five grey squares, which are not present in the grid. Hence, we have to start with a white square.

1. The largest number within the white squares is 7. If we start at 7, we can move to 8 and then to 3. From 3, if we go to 9, we have to choose between going left and going right. Any direction we pick, the other digits cannot be reached. Hence, from 3, we can go to 6. Then the remaining numbers are straightforward – 4, 9, 5, 1, 2.
Hence, the largest number that can be formed is 783649512. Ans: (783649512)

2. Given that the last 3 digits are 315. We can try first to form a nine-digit number ending with 315. We have already established that we have to start from a white square. If we start with 7, any direction that we travel, we have to use either 1 or 3 before reaching the last three digits.

Hence, 7 cannot be the starting number. Same is the case if we start with 4.
If we start with 2, we can form a nine-digit number 287649315. This is the largest number that Vanitha could have noted. Ans: (287649315)

3. Since the last digit must be a white square, any grey square and its adjacent white square can form the last two digits.
Hence, the possibilities are 15, 95, 12, 82, 87, 67, 64, 94, 83, 13, 63 and 93.
From the options, X can be divisible by 19 (for 95), can be divisible by 31 (for 93). But X cannot be divisible by 26. Hence, the answer is option B. Choice (B)

4. By observation, we can see that 7 cannot be a part of this number. This is because both the numbers that are adjacent to 7 differ by 1. Hence, the maximum number of digits in the number that Vanitha wrote has to be eight.
If we start from 2, the highest number that can be formed is 28364951 (with eight digits).
If we start from 8, we cannot go to 2 (since the next digit after 2 will be 1) or to 7 (since the next digit will be 6). Hence, the largest number will be 8364951, which is only seven digits.
Starting with 7, there are no possibilities. We need not check for 1, since any eight-digit number starting with 1 will be less than the number starting with 2 (28364951).
If we start with 3, we cannot form an eight-digit number.
If we start with 6, the highest number will be 64951382. This is higher than the number starting with 2.
We need not check for 4 and 5, since we have an eight-digit number starting with 6.
If we start with 9, we cannot form an eight-digit number.
Hence, the maximum possible number is 64951382. Ans: (64951382)

5. If all the nine-digits are present, then the sum of the digits will be 45. Hence, this number will be a multiple of 3.
For this to not be a multiple of 3, we need to leave out one number which is not 3 or 6 or 9.
Since there are only eight digits now, we can start with 9.
If we start with 9, the highest possible number is 95138764. This number is not divisible by 3 and is the highest possible number. The sum of the digits of this number is 43. Hence, statement IV is true.
All the other statements are false. Choice (D)

Solutions for questions 6 to 10:

Given that all the questions of type Q2 appeared only in Exam 2 (from (i)) and all the students attempted at least one question of QT2 (from (iv)). Hence, all the students must have attempted Exam 2.
The students could have attempted only Exam 2, only Exam 1 and Exam 2, only Exam 2 and Exam 3 or all the three exams. Let the number of students in these four groups be a, b, c and d, respectively.
From (i), $b + c = 27 \Rightarrow c = 27 - b$ (since no student attempted only Exam 3).
From (iii), $\frac{b+d}{c+d} = \frac{4}{11} = \frac{b+d}{27-b+d} = \frac{4}{11} \Rightarrow 15b + 7d = 108 \Rightarrow b = \frac{108-7d}{15}$
For both b and d to be positive, b = 3 and d = 9.
Hence, c = 24.
If QT1 is only in Exam 1, then 12 (i.e., b + d) students would have attempted this question type.
If QT1 is only in Exam 2 (or in Exam 2 and any other exam), then 36 + a (i.e., b + c + d + a) students would have attempted this question type.
If QT1 is only in Exam 3, then 33 (i.e., c + d) students would have attempted this question type.
If QT1 is only in Exam 1 and Exam 3, then 36 (i.e., b + c + d) students would have attempted this question type.
Same is the case for QT3.
From (ii), the ratio of the number of students who attempted at least one question of type QT1 to the number of students who attempted at least one question of type QT3 is 6:7.
Since QT1 was attempted by a lesser number of students than QT3, it cannot be in Exam 2 (since all the students attempted Exam 2).
Also, except for 36 + a, none of the other values are multiples of 7. Hence, QT3 must have been attempted by 36 + a students. This is equal to the number of students who took Exam 2, questions of the type QT3 must be in Exam 2.
The number of students who attempted QT1 must be a multiple of 6. Hence, this number can be 12 or 36. If the number of students who attempted QT1 is 12, then the number of students who attempted QT3 must be 14, which is not possible as a cannot be negative.
Hence, the number of students who attempted QT1 must be 36. The number of students who attempted QT3 must be 42 and a = 6.
Hence, QT1 is in Exam 1 and Exam 3;
QT2 is in Exam 2;
QT3 is in Exam 2 and may or may not be in each of Exam 1 and Exam 3.

The following table provides the number of students who attempted different combinations of exams:

Exams	Number of Students
Only Exam 2	6
Only Exam 1 and Exam 2	3
Only Exam 2 and Exam 3	24
Exam1, Exam 2 and Exam 3	9

6. The number of students who attempted only Exam 2 is 6. Ans: (6)

7. QT3 could have appeared in Exam 1, Exam 2 and Exam 3. Choice (D)

8. QT1 appeared in Exam 1 and Exam 3. Hence, 3 + 24 + 9 = 36 students must have attempted QT1. The answer is option B. Choice (B)

9. Total number of questions of type QT1 = Number of students who attempted Exam 1 or Exam 3 or both = 36
Total number of questions of type QT2 = 42 (which appeared only in Exam 2)
Total number of questions of type QT3 = 42 (since QT3 is in Exam 2 which was attempted by all the 42 students).
Hence, there must be a total of 120 questions.
From the given ratio, Exam 1 must have 24 questions, Exam 2 must have 60 questions and Exam 3 must have 36 questions.
Exam 2 must have 42 questions of type QT2 and the remaining 18 questions must be of type QT3.
Exam 1 and Exam 3 together must have 24 questions of type QT3.
Let the number of questions of type QT3 in Exam 3 be a. Then Exam 1 will have 24 – a questions of type QT3.
Since Exam 1 has a total of 24 questions, it must have a questions of type QT1.
Since Exam 3 has a total of 36 questions, it must have 36 – a questions of type QT1.
The maximum possible number of questions of type QT1 is 24. Choice (D)

10. From the above solution, the difference between the number of questions of type QT3 in Exam 1 and the number of questions of type QT1 in Exam 3 = (36 – a) – (24 – a) = 12.
Hence, the required difference is exactly 12. Choice (A)

Solutions for questions 11 to 15:

From the graph, we can see that, for College A, 18 experts gave a rating of at least 2. Hence, 25 – 18 = 7 experts must have given a rating of 1.
Also, 10 experts gave a rating of at least 3. Hence, 18 – 10 = 8 experts must have given a rating of exactly 2.
Let the number of experts who gave a rating of 5 to College A be a. From the given information, the number of experts who gave college A a rating of 1 was less than four-fifths but more than two-fifths the number of experts who gave the college a rating of 5.
Hence, $\frac{2a}{5} < 7 < \frac{4a}{5} \Rightarrow a < 17.5 < 2a \Rightarrow \frac{17.5}{2} < a \Rightarrow a > 8.75$
Hence, a minimum of 9 experts must have given College A a rating of 5.
Similarly, for College B, the number of experts who gave a rating of 1 was 25 – 19 = 6 and the number of experts who gave a rating of 2 was 19 – 12 = 7.
If b is the number of experts who gave a rating of 5 to College B, then
$\frac{2b}{5} < 6 < \frac{4b}{5} \Rightarrow b > 7.5.$
Hence, a minimum of 8 experts must have given College B a rating of 5.
For College C, the number of experts who gave a rating of 1 was 25 – 20 = 5 and the number of experts who gave a rating of 2 was 20 – 10 = 10.
If c is the number of experts who gave a rating of 5 to College C, then
$\frac{2c}{5} < 5 < \frac{4c}{5} \Rightarrow c > 6.25.$
Hence, a minimum of 7 experts must have given College C a rating of 5.
For College D, the number of experts who gave a rating of 1 was 25 – 17 = 8 and the number of experts who gave a rating of 2 was 17 – 12 = 5.
If d is the number of experts who gave a rating of 5 to College D, then
$\frac{2d}{5} < 8 < \frac{4d}{5} \Rightarrow d > 10.$
Hence, a minimum of 11 experts must have given College D a rating of 5.
For College E, the number of experts who gave a rating of 1 was 25 – 16 = 9 and the number of experts who gave a rating of 2 was 16 – 13 = 3.

If e is the number of experts who gave a rating of 5 to College E, then
$\frac{2e}{5} < 9 < \frac{4e}{5} \Rightarrow e > 11.25.$
Hence, a minimum of 12 experts must have given College E a rating of 5.

The following table provides the number of experts who rated each college from 1 to 5:

Rating	College A	College B	College C	College D	College E
1	7	6	5	8	9
2	8	7	10	5	3
3 or 4	0/1	0 to 4	0 to 3	0/1	0/1
5	10/9	12 to 8	10 to 7	12/11	13/12

11. The minimum number of experts who could have given a rating of 5 to College D is 11. Choice (A)

12. The maximum number of experts who could have given a rating of 4 to any of the five colleges = 1 + 4 + 3 + 1 + 1 = 10. Choice (A)

13. We can calculate the possible sum of ratings as it is indicative of the average as well.
Minimum sum of ratings of A = 7 + 16 + 3 + 45 = 71
Maximum sum of ratings of A = 7 + 16 + 0 + 50 = 73.
Minimum sum of ratings of B = 6 + 14 + 12 + 40 = 72.
Maximum sum of ratings of B = 6 + 14 + 0 + 60 = 80.
Minimum sum of ratings of C = 5 + 20 + 9 + 35 = 69.
Maximum sum of ratings of C = 5 + 20 + 0 + 50 = 75.
Minimum sum of ratings of D = 8 + 10 + 3 + 55 = 76.
Maximum sum of ratings of D = 8 + 10 + 0 + 60 = 78.
Minimum sum of ratings of E = 9 + 6 + 3 + 60 = 78.
Maximum sum of ratings of E = 9 + 6 + 0 + 65 = 80.
From the ranges above, we can see that the sum of the ratings (and hence, the average) cannot be the least for College D or College E.
Hence, College A, College B and College C can have the average rating as minimum. Choice (B)

14. The number of experts who gave a rating of either 1 or 2 for the five colleges A through E are 15, 13, 15, 13 and 12 respectively. Of these values, only for College E can it be equal to the number of experts who gave a rating of 5 to that college.
Hence, 12 experts gave a rating of 5 to College E. Choice (A)

15. From the above solution, the number of experts who gave a rating of 5 to College D cannot be 12 and must be 11.
For College B, the number of experts who gave a rating of 5 to College B cannot be either 11 or 12. Hence, it can only be 8 or 9 or 10.
The minimum possible average rating of College $B = \frac{6+14+12+40}{25} = 2.88$
The maximum possible average rating of College $B = \frac{6+14+8+50}{25} = 3.12$
Hence, the average rating is at least 2.88 and at most 3.12. Choice (C)

Solutions for questions 16 to 20:

The total cab fare of the six persons is 159 + 63 + 78 + 43 + 172 + 145 = ₹660.
Hence, the total amount with the six persons after they purchased pens will also be ₹660 (since the amount was redistributed, the total amount will remain the same).
The total initial amount the six persons had was 100 + 200 + 300 + 400 + 500 + 600 = ₹2100
The total amount spent on purchasing pens = 2100 – 660 = ₹1440
From (i), the cost of each pen was ₹45. Hence, the total number of pens purchased = 1440/45 = 32
From (ii), Ron had ₹185 after purchasing the pens. Hence, Ron must initially have had 300/400/500/600.

If Ron had 300, then Ron must have purchased pens for 300 – 185 = ₹115, which is not possible.
If Ron had 400, then Ron must have purchased pens for ₹215, which is also not possible.
If Ron had 500, then Ron must have purchased pens for ₹315, which is possible for 7 pens.
If Ron had 600, then Ron must have purchased pens for ₹415, which is again not possible.
Hence, Ron must have purchased 7 pens for ₹315 and he must have had ₹500 initially.

From (i), one person purchased 10 pens. This can only be the person with ₹500 or ₹600, since 10 pens cost ₹450. Since Ron had ₹500 and purchased only 7 pens, the person with ₹600 must have purchased 10 pens.
From (iv), Vin had half the amount that Tim had. Hence, Vin and Tim could have had ₹300 and ₹600 OR ₹200 and ₹400. Also, the number of pens that Vin purchased must be half the number of pens that Tim purchased. Only then, the amounts with them after the purchase of pens will be in the ratio 1:2.
If Vin had ₹300, then Vin must have purchased 5 pens (since Tim with ₹600 must have purchased 10 pens). Vin would have had 300 – 5 × 45 = ₹75 after purchasing the pens. Since Vin's cab fare is ₹172, the amount with him would have increased by less than ₹100, which violates condition (iii). Hence, Vin cannot have ₹300.
Therefore, Vin had ₹200 and Tim had ₹400. Since the amount with Vin after the purchase of pens must increase by more than ₹100 for him to pay his cab fare, the amount left with him must be less than ₹72 (since he paid ₹172 for his cab).
Hence, Vin must have purchased a minimum of 3 pens (which will leave him with 200 – 3 × 45 = ₹65) and a maximum of 4 (since he can only buy 4 pens with ₹200). From (v), he cannot buy 4 pens, since Saul purchased 4 pens. Hence, Vin must have purchased 3 pens and Tim must have purchased 6 pens.

Among the two persons who initially had ₹100 and ₹300, one must be Saul. Since Saul purchased 4 pens, Saul must be the person with ₹300.
The person with ₹100 must have purchased 2 pens (32 – 10 – 7 – 6 – 4 – 3). This person will have ₹10 immediately after purchasing the pens. Since Walt paid ₹145 for his cab and the amount with him decreased, Walt must be the person with ₹600 initially. Usain must have had ₹100 initially.

The following table provides, for the six persons, the initial amount with him, the number of pens purchased, the amount with him after purchasing the pen and the amount with him after redistribution (i.e., the cab fare):

Person	Amount	Pens	Amount Left	Cab Fare
Usain	100	2	10	43
Vin	200	3	65	172
Saul	300	4	120	63
Tim	400	6	130	78
Ron	500	7	185	159
Walt	600	10	150	145

16. Walt purchased 10 pens. Choice (A)

17. Usain had ₹10 after purchasing the pens. Ans: (10)

18. The amounts with only Usain and Vin increased after redistribution. Choice (B)

19. The total amount left with Saul and Tim combined = 120 + 130 = ₹250. Ans: (250)

20. The total number of pens purchased by Ron and Vin combined = 7 + 3 = 10. Choice (C)

Difficulty level wise summary - Section II

Level of Difficulty	Questions
Very Easy	–
Easy	2, 11
Medium	12, 13, 14, 15, 16, 17, 18, 19, 20
Difficult	1, 3, 4, 5, 6, 7, 8
Very Difficult	9, 10

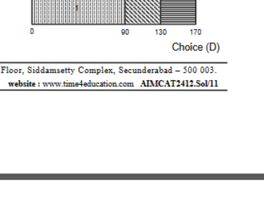
SECTION – III

Solutions for questions 1 to 22:

1. If the roots are real, then discriminant ≥ 0
$\Rightarrow p^2 - 4 \times 12 \geq 0 \Rightarrow p^2 \geq 48 \Rightarrow |p| \geq \sqrt{48}$
$\Rightarrow |p| \geq 4\sqrt{3}$
But p is the sum of the roots α_1 & α_2
$\Rightarrow |\alpha_1 + \alpha_2| \geq 4\sqrt{3}$ Choice (C)

2. $Q = 0.\,P_1 P_2 P_3 P_1 P_2 P_3 \ldots$
$1000Q = P_1P_2P_3 . P_1P_2P_3P_1P_2P_3 \ldots$
$999Q = P_1P_2P_3$
⇒ whenever Q is multiplied by 999 or multiples of 999, we get an integer among the choices, only 1998 is a multiple of 999. Choice (A)

3. Six tiles of 90 cm × 40 cm can be placed on the floor of dimensions 170 cm × 140 cm in the following way



Choice (D)

4. Since can observe that the only logarithm in the given equation is to the base 5, let us take $x = 5^a$
$\Rightarrow \log_5 12 + 3\log_x 16 = \log_5 12 + 3\log_{5^a} 16$
$= \log_5 12 + \frac{3}{a}\log_5 16$
$= \log_5 12 + \log_5 16^{\frac{3}{a}}$
$= \log_5[(12) \times (16^{\frac{3}{a}})]$
$\Rightarrow 25^{[\log_5 12 + 3\log_x 16]} = 5^{2\log_5(12)(16)^{\frac{3}{a}}}$
$= 5^{\log_5[(12)(16)^{\frac{3}{a}}]^2}$
$= [(12)(16)^{\frac{3}{a}}]^2$
$\Rightarrow [(12)(16)^{\frac{3}{a}}]^2 = \frac{9}{256}$
$\Rightarrow (12)(16)^{\frac{3}{a}} = \pm\frac{3}{16}$
But $(12)(16)^{\frac{3}{a}}$ cannot be negative.
$\Rightarrow (12)(16^{\frac{3}{a}}) = \frac{3}{16}$
$\Rightarrow 16^{\frac{3}{a}} = \frac{1}{64}$
$\Rightarrow 2^{\frac{12}{a}} = 2^{-6}$
$\Rightarrow a = -2$
∴ The value of $x = 5^{-2} = 0.04$

Alternative Solution:

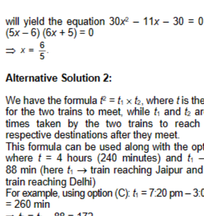
The given equation can be written as
$(5^2)^{[\log_5 12 + 3\log_x 16]} = \frac{9}{256}$
$\Rightarrow 2\log_5 12 + 6\log_x 16 = \log_5 9 - 2\log_5 16$ (taking log to base 5 on both sides)
$\Rightarrow 6\log_x(2^4) = \log_5(3^2) - 2\log_5(2^4) - 2\log_5(2 \times 3)$
$\Rightarrow 24\log_x 2 = \log_5\left(\frac{3^2}{2^8\, 2^4\, 3^2}\right)$
$\Rightarrow 24\log_x 2 = \log_5(2^{-12})$
$\Rightarrow 2\log_x 2 = -\log_5 2$
$\Rightarrow \frac{2\log 2}{\log x} = \frac{-\log 2}{\log 5}$
$\Rightarrow \log x = \log 5^{-2}$
$\Rightarrow x = 5^{-2}$ Ans: (0.04)

5. As (11 men + 16 boys) take 2 days
⇒ (22 men + 32 boys) take 1 day → (1)
Also, as (5 men + 11 boys) take 4 days
⇒ (20 men + 44 boys) take 1 day → (2)
∴ 22 men + 32 boys = 20 men + 44 boys ⇒ 1m = 6b.
Hence, 11 men and 16 boys work = 82 boys work and 1 man and 4 boys work = 10 boys work.
∴ Time taken by 1 man and 4 boys to complete the work $= \frac{82 \times 2}{10} = 16\frac{2}{5}$ Choice (A)

6. $|\sin\theta + \cos\theta| = \sqrt{2}\left|\frac{1}{\sqrt{2}}\sin\theta + \frac{1}{\sqrt{2}}\cos\theta\right|$
$= \sqrt{2}\left|\sin\left(\theta + \frac{\pi}{4}\right)\right| < \frac{1}{\sqrt{2}} \Rightarrow \left|\sin\left(\theta + \frac{\pi}{4}\right)\right| < \frac{1}{2}$
$\Rightarrow \frac{-1}{2} < \sin\left(\theta + \frac{\pi}{4}\right) < \frac{1}{2}$
Given $0 \leq \theta \leq 2\pi \Rightarrow \frac{\pi}{4} \leq \theta + \frac{\pi}{4} \leq 2\pi + \frac{\pi}{4}$
Let $\alpha = \theta + \frac{\pi}{4} \Rightarrow \frac{\pi}{4} \leq \alpha \leq \frac{9\pi}{4}$ and $\frac{-1}{2} < \sin\alpha < \frac{1}{2}$
$\Rightarrow \alpha \in \left(\frac{5\pi}{6}, \frac{7\pi}{6}\right) \cup \left(\frac{11\pi}{6}, \frac{13\pi}{6}\right)$
$\Rightarrow \theta \in \left(\frac{5\pi}{6} - \frac{\pi}{4}, \frac{7\pi}{6} - \frac{\pi}{4}\right) \cup \left(\frac{11\pi}{6} - \frac{\pi}{4}, \frac{13\pi}{6} - \frac{\pi}{4}\right)$
$\Rightarrow \theta \in \left(\frac{7\pi}{12}, \frac{11\pi}{12}\right) \cup \left(\frac{19\pi}{12}, \frac{23\pi}{12}\right)$
Of the given options only B satisfies. Choice (B)

7. Let the cost price per ml of milk be 1
CP of 1000ml = ₹1000
Let the quantity of water added to each 1000 ml of milk be U ml.
Now he sells (1000 + U) at ₹$\frac{4}{3}$ per ml to get ₹1400
$(1000 + U)\frac{4}{3} = 1400$
⇒ 4000 + 4U = 4200
⇒ U = 50 Ans: (50)

8. Let the speeds of the trains starting from Delhi and Jaipur be u_A and u_P respectively. Let us denote Delhi, Jaipur and their meeting points by A, P and X respectively.
Considering the train starting from Jaipur take t hours more to reach its destination after crossing the other train



Before meeting: $AX = u_A \times 4$, $PX = u_P \times 4$
After meeting: $XP = u_A\left(t + \frac{88}{60}\right)$, $XA = u_P(t)$
$(AX)(PX) = u_A u_P 16$
$(XP)(XA) = u_A u_P t\left(t + \frac{88}{60}\right)$
$\therefore u_A u_P 16 = u_A u_P t\left(t + \frac{88}{60}\right)$

$t\left(t + \frac{88}{60}\right) = 16$
$t^2 + \frac{22}{15}t = 16$
$15t^2 + 22t - 240 = 0$
$15t^2 + 72t - 50t - 240 = 0$
$3t(5t + 24) - 10(5t + 24) = 0$
$(3t - 10)(5t + 24) = 0$
$\therefore t = 10/3$
$\therefore t + \frac{88}{60} = \frac{10}{3} + \frac{22}{15} = \frac{72}{15} = \frac{24}{5}$ hrs.
Thus the train reaches Jaipur $\frac{24}{5}$ hrs after 3 pm. i.e., at 7:48 pm

Alternative Solution 1:

After the two trains met, the time taken by the train from Delhi to reach Jaipur $= 4\frac{V_P}{V_a}$ and the time taken by the train from Jaipur to reach Delhi $= 4\frac{V_a}{V_P}$
The difference of the above times is given as 88 minutes i.e., $4 \times 60\left(\frac{V_P}{V_a} - \frac{V_a}{V_P}\right) = 88.$
Let $\frac{V_P}{V_a} = x$.
$\therefore 240\left(x - \frac{1}{x}\right) = 88,$
i.e., $x - \frac{1}{x} = \frac{11}{30}$
Now, the above equation will have roots that are reciprocals of each other. Further, neither x nor $\frac{1}{x}$ can be greater than 2 (since the difference of reciprocals would then exceed 1 and we need the difference to be $\frac{11}{30}$).
Also, the LCM of the denominators is 30 (or a multiple of 30)
By trial and error we could look at $\frac{2}{15}, \frac{3}{10}, \frac{5}{6}$ out of which only $\frac{5}{6}$ satisfies the condition that both $\frac{5}{6}$ and $\frac{6}{5}$ are less than 2.
Hence, $\frac{6}{5} - \frac{5}{6} = \frac{11}{30}$ which satisfies $\Rightarrow \frac{V_P}{V_a} = \frac{6}{5}$
∴ The train from Delhi reached Jaipur at 3:00 pm + 4 hrs × $\frac{6}{5}$ = 3:00 pm + 240 × $\frac{6}{5}$ min = 3:00 pm + 288 min i.e., 7:48 pm.
Note: The equation $x - \frac{1}{x} = \frac{11}{30}$ could also be solved in the traditional way as a quadratic, which will yield the equation $30x^2 - 11x - 30 = 0$, i.e., $(5x - 6)(6x + 5) = 0$
$\Rightarrow x = \frac{6}{5}$

Alternative Solution 2:

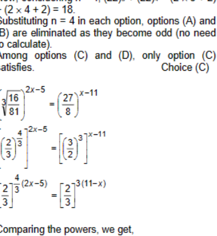
We have the formula $t^2 = t_1 \times t_2$, where t is the time for the two trains to meet, while t_1 and t_2 are the times taken by the two trains to reach their respective destinations after they meet.
This formula can be used along with the options, where t = 4 hours (240 minutes) and $t_1 - t_2$ = 88 min (here t_1 → train reaching Jaipur and t_2 → train reaching Delhi)
For example, using option (C): t_1 = 7:20 pm – 3:00 pm = 260 min
⇒ $t_2 = t_1 - 88 = 172$.
Now $(240)^2 \neq (260) \times (172)$
In this manner, we check for each option, and see that only option (A) satisfies. Choice (A)

9. Any natural number, A, which lies in the interval $1 \leq A \leq 36$ is capable of being the area of the rectangle formed by considering one or more of the playing squares vertically or horizontally provided it satisfies the following condition.
A = m × n, where m, n ∈ N and 1 ≤ (m, n) ≤ 6
∴ The value that A can take are 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 16, 18, 20, 24, 25, 30 and 36. A can take 18 distinct values.

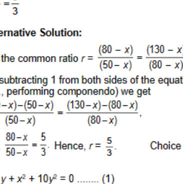
Alternative Solution:

Area of a rectangle is ℓ × b where ℓ and b are from 1 to 6.
Number of such rectangles which are not squares is $^6C_2 = 15$.
Number of squares = 6
∴ Total number of possible rectangles = 15 + 6 = 21.
But of these, 1 × 6 = 2 × 3
2 × 6 = 3 × 4
1 × 4 = 2 × 2
∴ There are 3 possible areas, each of which can be obtained from two distinct rectangles.
∴ Number of distinct areas possible = 21 – 3 = 18. Ans: (18)

10.



By joining the centres of the four corner most spheres, we get the following figure.



ACE is a right angled triangle, where E is the centroid of ΔBCD.
BCD is an equilateral triangle with the side equal to 6R, where R is the radius of each sphere.
⇒ CE = circumradius of the triangle BCD
$= \frac{6R}{\sqrt{3}} = 2\sqrt{3}R$
$AE^2 + EC^2 = AC^2$
$AE^2 + (2\sqrt{3}R)^2 = (6R)^2$
$\therefore AE^2 = 36R^2 - 12R^2 = 24R^2$
$\therefore AE = 2\sqrt{6}R$
∴ The height of the tetrahedron = The distance of the topmost point from the surface
$= R + AE + R = R + 2\sqrt{6}R + R$
$= 2(1 + \sqrt{6})R$ Choice (C)

11. It is given that 50 – x, 80 – x and (130 – x) are in geometric progression.
$\therefore (80 - x)^2 = (50 - x)(130 - x)$
$6400 - 160x + x^2 = 6500 - 180x + x^2$
⇒ 20x = 100
∴ x = 5
Therefore, the numbers in geometric progression are 45, 75 and 125 and the common ratio is $\frac{75}{45} = \frac{5}{3}$

Alternative Solution:

Let the common ratio $r = \frac{(80-x)}{(50-x)} = \frac{(130-x)}{(80-x)}$.
By subtracting 1 from both sides of the equation (i.e., performing componendo) we get
$\frac{(80-x)-(50-x)}{(50-x)} = \frac{(130-x)-(80-x)}{(80-x)}$
i.e., $\frac{80-x}{50-x} = \frac{5}{3}$. Hence, $r = \frac{5}{3}$. Choice (B)

12. $x + y + x^2 + 10y^2 = 0$ .... (1)
$\frac{x}{y} + \frac{y}{x} = 9$
$\Rightarrow x^2 + y^2 = 9xy$ ...... (2)
∴ (1) becomes $x + y + 9xy + 9y^2 = 0$
$(x + y)(1 + 9y) = 0$
As $x + y \neq 0$, $y = -\frac{1}{9}$
$\therefore 27y = -3.$ Ans: (–3)

13. $(22)_3 + (22)_4 + (22)_5 + \ldots + (22)_n =$
$= (2(3) + 2) + (2(4) + 2) + (2(5) + 2) + \ldots + (2(n) + 2)$
$= 2(3 + 4 + \ldots n) + 2(n - 2)$
$= 2(1 + 2 + 3 + 4 + \ldots n - (1 + 2)) + 2(n - 2)$
$= 2\left(\frac{(n)(n+1)}{2} - 3\right) + 2n - 4$
$= n^2 + n - 6 + 2n - 4$
$= n^2 + 3n - 10$

Alternative Solution:

Considering only the first term, i.e., $(22)_3$, n = 3 and $(22)_3 = 3 \times 2 + 2 = 8.$
Substituting n = 3 in each option, option (D) is eliminated.
Now, considering n = 4, $(22)_3 + (22)_4 = (2 \times 3 + 2) + (2 \times 4 + 2) = 18.$
Substituting n = 4 in each option, options (A) and (B) are eliminated as they become odd (no need to calculate).
Among options (C) and (D), only option (C) satisfies. Choice (C)

14. $\left(\sqrt[3]{\frac{16}{81}}\right)^{2x-5} = \left(\frac{27}{8}\right)^{x-11}$
$\left[\left(\frac{2}{3}\right)^{\frac{4}{3}}\right]^{2x-5} = \left[\left(\frac{3}{2}\right)^3\right]^{x-11}$
$\left[\frac{2}{3}\right]^{\frac{4}{3}(2x-5)} = \left[\frac{2}{3}\right]^{3(11-x)}$
Comparing the powers, we get,
$\frac{4}{3}(2x - 5) = 3(11 - x)$
⇒ 8x – 20 = 99 – 9x
⇒ x = 7 Choice (B)

15. The probabilities for 1, 2, 3, 4, 5 and 6 turning up are all equal. Hence for every six throws, the expected amount earned by A = 1 + 2 + 3 + 4 + 5 + 6 = 21
⇒ Average expected amount per throw $= \frac{21}{6} = 3.5$
∴ Expected net earnings of A = 3.5 – 3 = 0.5
For 300, throws expected net earnings = 300 × 0.5 = ₹150. Choice (A)

16. Since, f(x.y) = f(x).f(y)
We get
f(1.y) = f(1) f(y)
⇒ f(1) = 1
Also,
f(1).f(2).f(4).f(8).f(16).f(32).f(64)
= f(1).[f(64).f(2)].[f(32).f(4)].[f(16).f(8)]
= f(1).f(128).f(128).f(128) = 1 × 8 × 8 × 8 = 512

Alternative Solution:

If f(x.y) = f(x).f(y), then we can consider $f(x) = x^n$
$f(128) = 128^n = 8$
$2^{7n} = 2^3$
n = 3/7
$\therefore f(x) = x^{3/7}$
f(1) f(2) f(4).....f(64)
$= 1 \times 2^{3/7} \cdot 4^{3/7} \cdot 8^{3/7} \cdot 16^{3/7} \cdot 32^{3/7} \cdot 64^{3/7} = 512$ Ans: (512)

17. Let us denote the number as N
N = 31K + 3
K can be of the form 3a or 3a + 1 or 3a + 2.
∴ N = 31(3a) + 3 or 31(3a + 1) + 3 or 31(3a + 2) + 3.
∴ N = 93a + 3 or N = 93a + 34 or N = 93a + 65.
Therefore, when N is divided by 93, the remainder can be 3 or 34 or 65
Thus, three possible values exist for r Choice (D)

18. The number can begin with

1st digit	Remaining digits	No. of possibilities
4	0s → Seven	1
3	0s → Six 1s → One	$\frac{7!}{6!} = 7$
2	(i) 0s → Six 2s → One	$\frac{7!}{6!} = 7$
	(ii) 0s → Five 1s → Two	$\frac{7!}{5!\,2!} = 21$
1	(i) 0s → Six 3s → One	$\frac{7!}{6!} = 7$
	(ii) 0s → Five 1s → One	$\frac{7!}{5!} = 42$
	(iii) 0s → Four 1s → Three	$\frac{7!}{4!\,3!} = 35$

∴ Total number of numbers = 1 + 7 + 7 + 21 + 7 + 42 + 35 = 120 Ans: (120)

19. Difference in compound interest and simple interest for two years = Interest on the simple interest for the first year.
S.I for 2 years = 2 × S.I for 1 year = 720
⇒ (S.I) for 1 year = 360
⇒ 792 – 720 = $\frac{360 \times 1 \times R}{100}$
⇒ 72 = $\frac{360 \times R}{100}$
⇒ R = 20%
∴ Rate of Interest = 20% per annum Choice (C)

20. $^{100}C_{50} = \frac{100!}{50!\,50!}$
To find the number of trailing zeroes in $^{100}C_{50}$, we need to find the number of trailing zeroes in 100! and those in 50!
Highest power of 5 in 50! $= \left[\frac{50}{5}\right] + \left[\frac{50}{5^2}\right] = 10 + 2 = 12$
The highest power of 2 in 50! will be definitely greater than that of 5.
Thus 50! has 12 trailing zeroes
∴(50!)(50!) will have 24 trailing zeroes.
Proceeding similarly, we can find 100! will also have 24 trialing zeroes.
$\therefore \frac{100!}{(50!)(50!)}$ will have no trailing zeroes.

Alternative Solution:

The number of trailing zeroes in any factorial depends on the number of 5s appearing in that factorial.
It can be observed that the number of times 5 appears in 1 × 2 × 3 × 4 × ..... × 49 × 50 is same as that in 51 × 52 × ..... × 100 (both products containing 10 multiples of 5, 2 of which are also multiples of 25).
Hence, the number of 5s in 100! will be equal to the number of 5s in 50! 50!
⇒ $\frac{100!}{50!\,50!}$ will have zero trailing zeroes. Ans: (0)

21. Since $S_m = \frac{m(m+1)}{2}$ must be a multiple of 15.
We need m(m + 1) = multiple of 2 × 15 = 2 × 3 × 5
i.e., m(m + 1) must contain a factor of 2, 3 and 5.
Now, m(m + 1) is always even.
⇒ 2 is always a factor of m(m + 1).
Now, consider all multiples of 5 less than or equal to 100 (i.e., a total of 20 multiples). Since among every three consecutive numbers there is always one multiple of three, for all multiples of 5, it will be possible to find at least one value of m (or m + 1) such that m(m + 1) contains a factor of 3 (i.e., apart from containing a factor of 5). Hence at least 20 values of m are there.
However, in the above approach, if in any of the cases, the multiple of 5 itself is also a multiple of 3 then one more value of m will possible.
Hence, for all the multiples of 15 we get an extra possible value of m, i.e., 6 multiples of 15 below 100
⇒ another six values of m are possible.
Hence, a total of 26 values are possible. Ans: (26)

22. Let the number of students be x.
x × 45 + 15 × 60 = (x + 15) × 50
⇒ 45x + 900 = 50x + 750
⇒ 5x = 150
∴ x = 30 Choice (D)

Difficulty level wise summary - Section III

Level of Difficulty	Questions
Very Easy	
Easy	5, 7, 14, 19, 22
Medium	1, 2, 3, 4, 11, 13, 15
Difficult	6, 8, 9, 12, 16, 17, 18, 20, 21
Very Difficult	10

Done
Tip: Scroll slowly for the best results